Lewis Liu

AI Engineer · LLM Agents, Self-Hosted LLMs & Developer Tools

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Open to AI application and agent engineering, AI automation and internal tools, and full-stack roles · Based in mainland China, open to relocation and remote

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SUMMARY

- Software engineer with four years across LLM agents, on-prem model serving, and production genomics pipelines.
- Shipped open-source BioinfoFlow, Trove, Loadout, and Folio.
- M.S. in Molecular Biology with genomic and clinical data experience.

OPEN-SOURCE PROJECTS

BioinfoFlow Dec 2025 – Present

Agentic bioinformatics workflow platform · MIT · Side project · bioinfoflow.com · GitHub · Python, TypeScript, FastAPI, Next.js, Docker

- Shipped a natural-language bioinformatics platform: the agent reads the project, prepares inputs and parameters, submits Nextflow or WDL runs after user approval, follows logs, and explains results.
- Kept data and compute on the user's own hardware with a local-first architecture.
- Architected the agent runtime for resumable long-running tasks, using durable sessions, append-only history, and a small tool set (`read` , `bash` , `edit` , `write` , `ask_user`) that drives the product through its CLI.
- Added approval gates for risky actions through three permission modes: ask before changes, ask before dangerous actions, and full access.
- Implemented context compression, retries, and cancellation to keep long sessions stable.
- Built a unified execution backend that drives Nextflow and MiniWDL through one interface, using FastAPI and a Typer CLI.
- Built a persistent scheduler for queueing, resource checks, retries, timeouts, and cleanup, streaming run state over SSE and WebSocket.
- Delivered a Next.js workspace covering projects, dashboard, agent, workflows, runs (logs, DAGs, outputs), images, scheduler, and settings.
- Added SSH remote host management with connection checks, in-browser terminals, and per-host agent access, sandboxing remote commands in Bubblewrap.
- Released v0.1.0 and v0.2.0 (Jul 2026) with a one-line installer, packaged as Docker Compose.

- Wrote nf-core/rnaseq, WDL, and NVIDIA Parabricks WGS examples; GPU workflows run on a single consumer GPU such as an RTX 4080 SUPER.

Trove Nov 2026

Workspace for starred GitHub repositories · MIT · GitHub · TypeScript, Next.js, React, SQLite

- Organizes starred GitHub repos with tags, collections, and AI categorization, stored in local SQLite.
- Maps related repos in a force-directed graph linked by shared tags, collections, and topics.
- Surfaces new repos through Radar, drawing on GitHub search and Hacker News.
- Tracks new releases of repos in use and flags likely breaking changes.
- Generates AI health reports for repositories.
- Exposes a built-in MCP server to coding assistants via tokens minted and revoked from the CLI.
- Supports OpenRouter, OpenAI, Anthropic, DeepSeek, local Ollama, and any OpenAI-compatible server, with Ollama restricted to loopback addresses.
- Ships an English and Chinese UI and runs with Docker Compose.

Loadout Oct 2026

macOS manager for AI agent skills · MIT · GitHub · Rust, Tauri 2, React, SQLite

- Shipped v0.2.1 as a universal build for Apple silicon and Intel Macs.
- Unified skills from `~/.agents`, Claude Code, Codex, Cursor, Gemini, and OpenCode into one deduplicated list, showing each symlinked skill once with all its install locations.
- Built in-app install, update, and delete, with automatic backups before updates and deletes sent to the Trash.
- Built a skill-usage heatmap from agent session logs that stores only skill, agent, and day, using a Rust scanner, debounced file watcher, log parsers, and an incremental SQLite index.
- Delivered an English and Chinese UI with light, dark, and system themes.
- Limited file access to the user's home folder, with no Full Disk Access, Automation, Accessibility, or Screen Recording permission.

Folio Nov 2026

Local reading library for saved X Articles · Apache-2.0 · macOS Apple Silicon preview · GitHub · TypeScript, Electron, React

- Built a Chrome extension that imports X Articles from bookmarks in the background, without opening or refreshing X tabs.
- Stored articles and images as local Markdown, so the same folder opens in Obsidian or Cursor.
- Added browsing by topic and reading status, ratings, and annotations.
- Added bring-your-own-key AI that turns selected articles into topic notes and cards for review before saving, with keys encrypted in the system's secure storage.
- Delivered an English and Chinese UI with light, dark, and system themes.

EXPERIENCE

BGI Genomics · Software Development Engineer, Shenzhen Jul 2022 – Oct 2026

Self-hosted LLM service Jun – Aug 2026

- Designed and launched a fully on-prem LLM service behind an OpenAI-compatible API for the team knowledge base, BioinfoFlow, and internal projects, serving DeepSeek-V4-Flash on 8×H20 via vLLM (tensor + expert parallelism, FP8 KV cache).
- Pinned the vLLM image by digest to prevent version drift on re-pulls.
- Built a multi-tenant LiteLLM gateway that issues virtual keys per colleague, project, and service, keeping the raw model key private.
- Enforced per-key limits on requests per minute, tokens per minute, budget, and parallel calls.
- Shipped key management scripts to create, inspect, block, unblock, and rotate keys.
- Built GPU and inference observability with Prometheus, DCGM exporter, and Grafana.
- Designed a capacity dashboard tracking health, estimated headroom, running and waiting requests, KV cache usage, TTFT P95, and TPOT P95.
- Designed a usage dashboard reporting requests and tokens per key.
- Fixed misaligned metrics by tracing an 8-hour server clock skew, adding a query offset, and documenting the rollback once NTP is fixed.
- Deployed BioinfoFlow v0.2.0 on the same server, sharing the LiteLLM gateway with OpenWebUI.
- Delivered reusable agent skills for sequencing workflows, WGS, and platform operations.
- Wrote the operations runbook covering start and stop, key management, dashboards, and troubleshooting.

OpenWebUI knowledge base Jun – Aug 2026

- Launched an on-prem OpenWebUI knowledge base (v0.9.6) that keeps chat, embeddings, file parsing, and vector search inside the network.
- Designed the storage and parsing stack: PostgreSQL for users, chats, and permissions; PGVector for embeddings; Redis for sessions, token revocation, and WebSocket; Apache Tika for Word, PDF, PowerPoint, and Excel.
- Deployed a local BGE-M3 embedding service (1024 dimensions) on the DeepSeek server, lowering DeepSeek's GPU memory share from 0.90 to 0.85 to free capacity.
- Enabled hybrid search and set starting values for chunk size, overlap, and top-k.
- Wrote the production deployment guide covering Docker Compose setup, verification, backup and restore, upgrades, scale-out criteria, and common failures.
- Defined the launch checklist, including a recall test on real business questions and a stability check under multi-user load.

Medical LLM evaluation Dec 2023 – Feb 2024

- Built an LLM pipeline that extracts ACMG evidence criteria from variant data to support clinical variant interpretation.
- Built a prompt evaluation benchmark, medprompt-eval.

Domain fine-tuning

- Fine-tuned LLaMA-7B and BioGPT on ~10,000 HPO records, including data cleaning, instruction building, training, and evaluation.

Production NGS pipelines

- Developed and maintained 10+ production NGS pipelines (pan-cancer, NIPT, WGS, hereditary hearing loss) covering QC, alignment, variant calling, annotation, and reporting.
- Introduced pipeline engineering standards: Git versioning, input/output contracts, containerized runtimes, and unit tests.
- Ran and debugged jobs on HPC clusters.
- Documented edge cases, parameter limits, and failure modes to make production runs more stable and reproducible.

SKILLS

Area	Skills
AI / LLM	Agent runtimes and tool calling, prompt engineering, LLM evaluation and benchmark design, fine-tuning, training data, RAG; Claude Code, Codex, Cursor
LLM serving	vLLM, LiteLLM, OpenWebUI, PGVector, BGE-M3, Apache Tika, Prometheus, Grafana, DCGM
Languages	Python, TypeScript, Rust, Bash
Backend and data	FastAPI, Typer, SSE / WebSocket, PostgreSQL, SQLite, Redis
Frontend and desktop	Next.js, React, Tailwind CSS, Tauri 2, Electron
Infrastructure	Docker / Compose, Linux, Git, SSH, HPC, NVIDIA GPU
Workflow and domain	Nextflow, nf-core, WDL, Snakemake, GATK4, Parabricks; genomic and clinical variant data
English	CET-6

AWARDS

- Champion and Innovation Award, Life Science track, Alibaba Cloud Tianchi World Science Intelligence Competition (2023): predicted biological age and disease risk from DNA methylation data with AutoGluon and MLP models.

EDUCATION

- South China Agricultural University, B.S. and M.S. in Molecular Biology, 2015 – 2022.

PUBLICATION

- Co-author, "Fast and accurate DNASeq variant calling workflow composed of LUSH toolkit," *Human Genomics*, 2024.